

- **File opt:** Lets you enable/disable recovery of certain file types.

After selecting the 'Search' menu item for a particular partition, the user is asked whether to recover the whole partition, or only unallocated space. The first option is generally selected when the filesystem gets corrupted, while the latter is selected when the underlying filesystem is intact, and the user wants to recover only the deleted files.

Last, the user has to choose the output directory used for writing recovered files. After this, recovery begins; you will see a screen showing the progress of the recovery. Once completed, a summary is displayed. If interrupted, the recovery process can be resumed at the next start-up of PhotoRec. To learn more, refer to:

- <http://www.howtogeek.com/howto/15761/recover-data-like-a-forensics-expert-using-an-ubuntu-live-cd/>
- <http://www.linux.com/archive/feed/56588>

TestDisk

TestDisk is a powerful data-recovery tool, which recovers lost data partitions and/or makes non-bootable disks bootable again. Such cases can occur due to faulty/malicious software, certain viruses, or due to human error (such as accidentally erasing the partition). TestDisk supports all popular partition table types, including the widely used PC/Intel partition table, Apple partition map, GUID partition table and Sun Solaris slices. Also, it has support for most filesystems, including Windows FAT/NTFS, Linux ext2/ext3/ext4, Linux RAID, Linux swap and Apple HFS/HFS+/HFSX.

TestDisk starts by asking the user for an optional log file for the details of the recovery process. After this, the user is asked to select the appropriate media, and the type of the partition table. Next, the user is provided with a menu like that in Figure 2.

- Analyse:** This lets the user analyse the current partition structure, and subsequently search for lost partitions. Each discovered partition is added to the list of found partitions, which are displayed after the end of the search process. One can use short-cuts such as 'P' to list files, and 'T' to change the partition type on any found partitions. The string 'Ok' appears at the bottom of this list; if the partition structure is correct, pressing 'Enter' will present options to either write the data into the partition table, or do a deeper search for partitions.
- Delete:** This will delete all partition data from the partition table. Both the MBR and the signature bytes (if any), remain the same.
- MBR code:** Selecting this option will overwrite the MBR code with a copy of a standard Master Boot Record. This option might be useful in

cases where the system fails to boot using the selected media.

- Disk geometry:** This option is an advanced one, and changes the disk geometry parameters (cylinders, head, sectors and sector size) for more comprehensive disk analysis. Based on these parameters only, TestDisk searches the partitions, and calculates their corresponding sizes. These parameters will not affect the hard disk until the user writes the data to the drive. Therefore, these parameters should be proper ones. The TestDisk utility, by default, gets these parameter values using BIOS calls under DOS, or specific system calls under Linux, Sun Solaris and Windows. If these default parameter values differ from those that are used while creating the partition table on the disk, warning messages will be displayed on the screen during the 'Analyse' phase. Hence, in such cases, this option allows tweaking of these parameters for the 'Analyse' phase.
- Options:** This allows the user to tweak certain options used in the 'Analyse' phase.
- Advanced:** It allows the user to fix the boot sector from a backup boot sector for a partition, to fix the MFT (Master File Table) for a partition, to image a particular partition, or to un-delete a partition.

To learn more, refer to:

- <http://bandwidthco.com/whitepapers/compforensics/datrecovery/testdisk/TestDisk%20Installation%20and%20Usage.pdf>
- http://www.cgsecurity.org/wiki/TestDisk_Step_By_Step

Data recovery/carving from media/images plays an important role in digital forensic analysis, as it helps in recovering data linked to a cyber crime. Also, the same set of tools can be helpful to those who have lost their data due to accidental deletion of files/directories, corruption of the filesystem holding the data file, or corruption/reformatting/deletion of the partition where the data resided. By exploring the various tools available for data recovery/carving from media/images in BackTrack, users will certainly find this Live Linux distribution very useful.

After data recovery comes the last part of a digital forensic investigation—comprehensive data analysis to find strong evidence related to the cyber crime. This will be covered in the next, and last, part of this series on digital forensics. 

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